Mumbai University Bachelor's of Computer Engineering (syllabus)

Semester V

Course code	Course name	Module Name
CSC501	Theoretical Computer Science	1.Basic Concepts and Finite Automata 2.Regular Expressions and Languages 3. Grammars 4.PushdownAutomata (PDA) 5. Turing Machine ™ 6. Undecidability
CSC502	Software Engineering	1. Introduction To Software Engineering and Process Models. 2. Software Requirements Analysis and Modeling 3. Software Estimation Metrics 4. Software Design 5. Software Testing 6. Software Configuration Management, Quality Assurance and Maintenance
CSC503	Computer Network	1. Introduction to Networking 2. Physical Layer 3. Data Link Layer 4. Network layer 5. Transport Layer 6. Application Layer
CSC504	Data Warehousing & Mining	1. Data Warehousing Fundamentals 2. Introduction to Data Mining, Data Exploration and Data Pre-processing 3. Classification 4. Clustering 5. Mining frequent patterns and associations 6. Web Mining
CSDLO501x	Department Level Optional Course- 1	

CSDLO501	Probabilistic Graphical Models	1. Introduction to Probabilistic Graphical Modeling 2. Bayesian Network Model and Inference 3. Markov Network Model and Inference 4.Hidden Markov Model and Inference 5. Learning and Taking Actions and Decisions 6. Applications
CSDLO501 CSDLO501	:Internet Programming	1. Introduction to Web Technology 2. Front End Development 3. Back End Development 4. Rich Internet Application (RIA) 5. Web Extension: PHP and XML 6. React js
	: Advance Database Management System	1. Distributed Databases 2. Distributed Database Handling 3. Data interoperability – XML and JSON 4. NoSQL Distribution Model 5. NoSQL using MongoDB 6. Trends in advance databases
CSL504	Professional Comm. & Ethics II	1. ADVANCED TECHNICAL WRITING: PROJECT/PROBLEM BASED LEARNING (PBL) 2. EMPLOYMENT SKILLS 3. BUSINESS MEETINGS 4. TECHNICAL/ BUSINESS PRESENTATIONS 5. INTERPERSONAL SKILLS 6. CORPORATE ETHICS

Semester VI

Course Code	Course Name	Module Name
CSC601	System Programming & Compiler Construction	1. Introduction to System Software 2. Assemblers 3. Macros and Macro Processor 4. Loaders and Linkers 5. Compilers: Analysis Phase 6. Compilers: Synthesis phase
CSC602	Cryptography & System Security	1. Introduction - Number Theory and Basic Cryptography 2. Symmetric and Asymmetric key Cryptography and key Management 3. Cryptographic Hash Functions 4. Authentication Protocols & Digital Signature Schemes 5. Network Security and Applications 6. System Security
CSC603	Mobile Computing	1. Introduction to Mobile Computing 2. GSM Mobile services 3. Mobile Networking 4. Wireless Local Area Networks 5. Mobility Management 6. Long-Term Evolution (LTE) of 3GPP
CSC604	Artificial Intelligence	1. Introduction to Artificial Intelligence 2. Intelligent Agents 3. Problem solving 4. Knowledge and Reasoning 5. Planning and Learning 6. AI Applications
CSDLO601x	Department Level Optional Course -2	
CSDLO6011	Internet of Things	1. Introduction to Internet of Things (IoT) 2. Things in IoT 3. The Core IoT Functional Stack 4. Application Protocols for IoT 5. Domain Specific IoTs 6. Create your own IoT
CSDLO6012	Digital Signal & Image Processing	1. Discrete-Time Signal and Discrete-Time System 2. Discrete Fourier Transform 3. Fast Fourier Transform 4. Digital Image Fundamentals 5. Image Enhancement in Spatial domain 6. Image Segmentation

CSDLO6013	Quantitative Analysis	1. Introduction to Statistics 2. Data Collection & Sampling Methods 3. Introduction to Regression 4. Introduction to Multiple Linear Regression 5. Statistical inference 6. Tests of hypotheses